The Actuarial AI Threshold: Can Large Language Models Pass IFoA Professional Examinations?

Abstract

This paper presents a systematic evaluation of the capability of a Large Language Model (LLM), Gemini 3.1 Pro in high-reasoning mode, to pass the professional examinations set by the Institute and Faculty of Actuaries (IFoA). In an experiment covering 28 exam papers from the September 2025 sitting — spanning Core Principles (CB, CM, CS), Core Practices (CP), Specialist Principles (SP), and Specialist Advanced (SA) subjects — the model achieved a 100% pass rate under all three grading conditions (Strict, Fair, and Generous). The average score under Fair grading was 87.6%, with an average margin of +30.1 percentage points above the published pass marks.

Both the exam-taking and grading phases were performed by the same model, introducing an inherent self-assessment bias that is partially mitigated by a tri-persona grading framework and selective human validation. Under the most punitive Strict grading, five exams passed with a margin of 5 percentage points or less. These caveats are discussed in detail alongside the headline results.

The findings provide strong evidence that modern LLMs possess the technical, mathematical, and commercial reasoning capabilities tested by the IFoA syllabus. However, exams are only one layer of the IFoA's qualification and quality framework, which also requires supervised work experience, ethical obligations, peer review, Practising Certificates for reserved roles, and Continuing Professional Development (CPD). This paper aims to prompt discussion within the profession on the role AI might play in actuarial work.

1. Introduction

The actuarial profession relies on a rigorous series of examinations administered by the Institute and Faculty of Actuaries (IFoA) to ensure that practitioners possess the mathematical, statistical, and commercial knowledge required to manage financial risk. These exams have historically stood as a high barrier to entry, testing not only technical knowledge but also the application of complex principles to novel scenarios.

The rapid advancement of Large Language Models (LLMs) has prompted a re-evaluation of cognitive and analytical work across many professions. While early AI models struggled with the exact precision required for advanced mathematics, recent reasoning-focused models have demonstrated significant improvements in logical reasoning, multi-step derivations, and professional-quality written analysis.

This experiment was conducted to systematically test whether a modern LLM could meet the technical standard set by the IFoA across the full breadth of its qualification syllabus. The study covers 28 exam papers spanning all four qualification stages, from foundational Core Principles through to the Specialist Advanced subjects. The aim is to provide clear, reproducible data on current AI capabilities within an actuarial context, with transparent discussion of both the strengths and the methodological limitations of the approach.

2. Methodology

2.1 Scope and Exam Selection

The experiment covered 28 complete exam papers from the **September 2025** sitting of the IFoA examinations. This sitting was specifically chosen because the papers were published after the model's training data cutoff (January 2025), eliminating the risk of data contamination (i.e., the model could not have memorised these questions during training).

The subjects covered span all four stages of the IFoA qualification:

Stage	Subjects	Papers
Core Principles	CB1, CB2, CM1 (A+B), CM2 (A+B), CS1 (A+B), CS2 (A+B)	10
Core Practices	CP1 (Papers 1+2), CP2 (Papers 1+2), CP3	5
Specialist Principles	SP1, SP2, SP4, SP5, SP6, SP7, SP8, SP9	8
Specialist Advanced	SA1, SA2, SA3, SA4, SA7	5

For subjects with separate Paper A (written/theoretical) and Paper B (practical/computer-based) components, both papers were attempted and graded independently.

2.2 Model Selection and Configuration

The model used for both the exam-taking and grading phases was **Gemini 3.1 Pro** in its high-reasoning configuration ("High" mode), which enables extended chain-of-thought processing before producing a response.

All experiments were run using Google Antigravity's standard chat mode with the "High" reasoning setting selected. No manual parameter overrides (temperature, token limits, etc.) were applied — the platform's default configuration was used throughout. The exact values of these defaults were not recorded and are not exposed by the Antigravity interface, which is a reproducibility limitation.

2.3 Experimental Architecture

The experiment was orchestrated within **Google Antigravity**, an agentic coding and automation environment. A dual-agent architecture was used, simulating the candidate-examiner dynamic:

1. **The Exam-Taker Agent:** Provided with the exam question paper and instructed to act as an actuarial candidate. It produced a structured answer document showing full working for every question. During the attempt phase, the agent had access only to:
    - The exam question paper (converted from PDF to Markdown).
    - The official IFoA Formulae and Tables book (also converted to Markdown).
    - Its own pre-trained knowledge (acting as the candidate's study and preparation).
    - Any provided Excel workbooks or R data files alongside the exam question paper.

    The agent was explicitly prohibited from accessing the examiner's report, marking scheme, or any solutions material.

2. **The Exam-Grader Agent:** A separate instantiation of the same model was subsequently provided with three inputs: the question paper, the LLM's attempt, and the official IFoA examiner's report. It evaluated the attempt against the published marking scheme, awarding marks question-by-question.

Important disclosure: Both the exam-taker and exam-grader agents used the same underlying model (Gemini 3.1 Pro High). While the two agents operated as independent sessions with no shared context, the use of the same model family for self-assessment introduces a potential bias, discussed further in Section 5 (Limitations).

2.4 Grading Framework: Tri-Persona Assessment

To account for the inherent subjectivity in marking discursive and applied answers, and to provide a confidence interval around the scores, the exam-grader evaluated each paper under three concurrent marking personas:

- **Strict (Harsh):** Marks awarded only when the answer directly matches the specific keywords, phrasing, or methodology in the examiner's report. No benefit of the doubt for vague explanations or alternative approaches. This represents a lower bound on performance.
- **Fair (Standard):** Marks awarded for answers that are mathematically equivalent, logically sound, or demonstrate clear understanding, even if the wording differs from the examiner's model answer. This is intended to approximate a human marker exercising professional judgement, and is treated as the **primary metric** throughout this paper.
- **Generous (Lenient):** Marks awarded for any valid, insightful point, including those that go beyond or take a different angle to the official marking scheme. This represents an upper bound.

Each question part was graded sequentially across all three personas before moving to the next part, ensuring consistent comparison. The grader was instructed to explicitly list the marking-scheme points awarded under each persona for auditability.

2.5 Data Preparation: PDF to Markdown Conversion

IFoA exam papers and examiner reports are published as PDF documents. These were converted to Markdown using a Python script (`pypdf` library) to enable clean text processing by the LLM.

This conversion is imperfect: mathematical notation, complex tables, and formatted formulae can be garbled during extraction. This is a known limitation that may have affected the model's interpretation of some questions and the grader's interpretation of some marking schemes, particularly for papers heavy in mathematical notation (CM, CS subjects). Where possible, the original PDFs were cross-referenced to verify accuracy.

2.6 Exam Execution Environment

A notable aspect of this experiment is that practical/computer-based exams were not merely answered theoretically. The exam-taker agent had access to a live execution environment:

- **Excel-based exams** (CM1B, CM2B, CP2): The agent read base data from provided `.xlsx` workbooks using Python (`openpyxl`), wrote computation scripts implementing year-by-year actuarial calculations (unit fund projections, profit vectors, zeroisation), and executed them to produce exact numerical results. Output workbooks were saved alongside the attempt.
- **R-based exams** (CS1B, CS2B): The agent wrote and executed R scripts to process provided `.RData` and `.csv` data files, performing statistical analyses (GLMs, time series, survival analysis, simulation) and generating plots.
- **Modelling exams** (CP2): The agent built complete Excel financial models programmatically, including audit trails, parameter sheets, and automated checks.

All scripts were executed and their outputs captured — the results reported in this paper are based on actual computed values, not theoretical estimates.

2.7 Human Validation

To partially mitigate the self-grading concern, selective human spot-checking of grade reports was conducted. This involved an actuary independently reviewing a sample of grade reports against the corresponding examiner reports to verify that marks were awarded appropriately. While not a full independent re-grading of all 28 papers, this provided a check on the grading agent's calibration.

3. Results

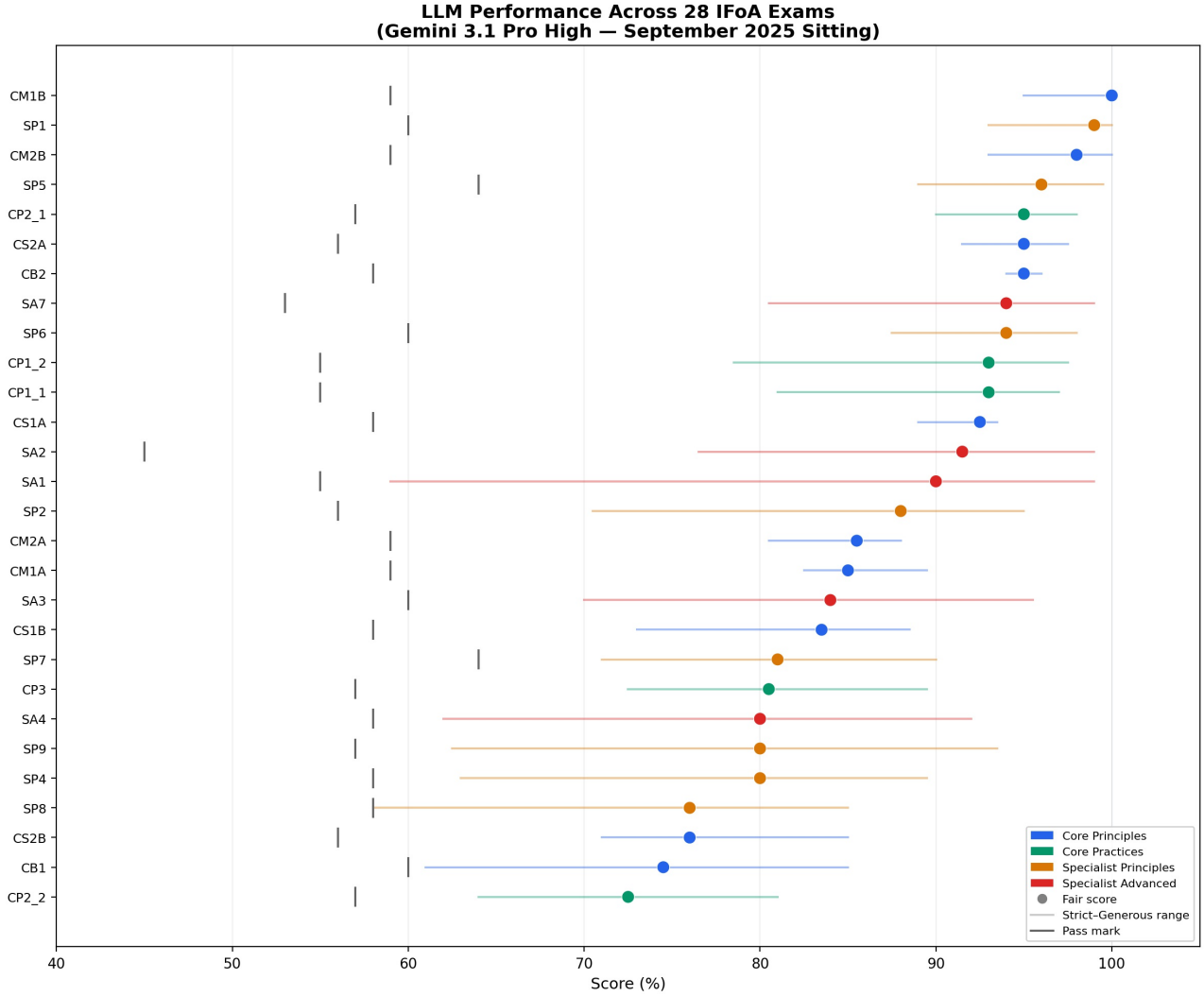
3.1 Overall Performance Summary

The model passed all 28 exams under all three grading conditions.

Metric	Strict	Fair	Generous
Exams Attempted	28	28	28
Exams Passed	28	28	28
Pass Rate	100.0%	**100.0%**	100.0%
Average Score	77.1%	**87.6%**	93.6%
Average Margin vs Pass Mark	+19.6%	**+30.1%**	+36.1%

3.2 Detailed Results by Exam

The table below shows individual exam results. Pass marks are set by the IFoA per sitting and vary by subject (ranging from 45% to 64% in this sitting). Figure 1 provides a visual overview.



Subject	Pass Mark	Strict	Fair	Generous	Strict Margin
CB1	60%	61.0%	74.5%	85.0%	+1.0%
CB2	58%	94.0%	95.0%	96.0%	+36.0%
CM1A	59%	82.5%	85.0%	89.5%	+23.5%
CM1B	59%	95.0%	100.0%	100.0%	+36.0%
CM2A	59%	80.5%	85.5%	88.0%	+21.5%
CM2B	59%	93.0%	98.0%	100.0%	+34.0%
CP1_1	55%	81.0%	93.0%	97.0%	+26.0%
CP1_2	55%	78.5%	93.0%	97.5%	+23.5%
CP2_1	57%	90.0%	95.0%	98.0%	+33.0%
CP2_2	57%	64.0%	72.5%	81.0%	+7.0%
CP3	57%	72.5%	80.5%	89.5%	+15.5%
CS1A	58%	89.0%	92.5%	93.5%	+31.0%
CS1B	58%	73.0%	83.5%	88.5%	+15.0%
CS2A	56%	91.5%	95.0%	97.5%	+35.5%
CS2B	56%	71.0%	76.0%	85.0%	+15.0%
SA1	55%	59.0%	90.0%	99.0%	+4.0%
SA2	45%	76.5%	91.5%	99.0%	+31.5%
SA3	60%	70.0%	84.0%	95.5%	+10.0%

SA4	58%	62.0%	80.0%	92.0%	+4.0%
SA7	53%	80.5%	94.0%	99.0%	+27.5%
SP1	60%	93.0%	99.0%	100.0%	+33.0%
SP2	56%	70.5%	88.0%	95.0%	+14.5%
SP4	58%	63.0%	80.0%	89.5%	+5.0%
SP5	64%	89.0%	96.0%	99.5%	+25.0%
SP6	60%	87.5%	94.0%	98.0%	+27.5%
SP7	64%	71.0%	81.0%	90.0%	+7.0%
SP8	58%	58.0%	76.0%	85.0%	+0.0%
SP9	57%	62.5%	80.0%	93.5%	+5.5%

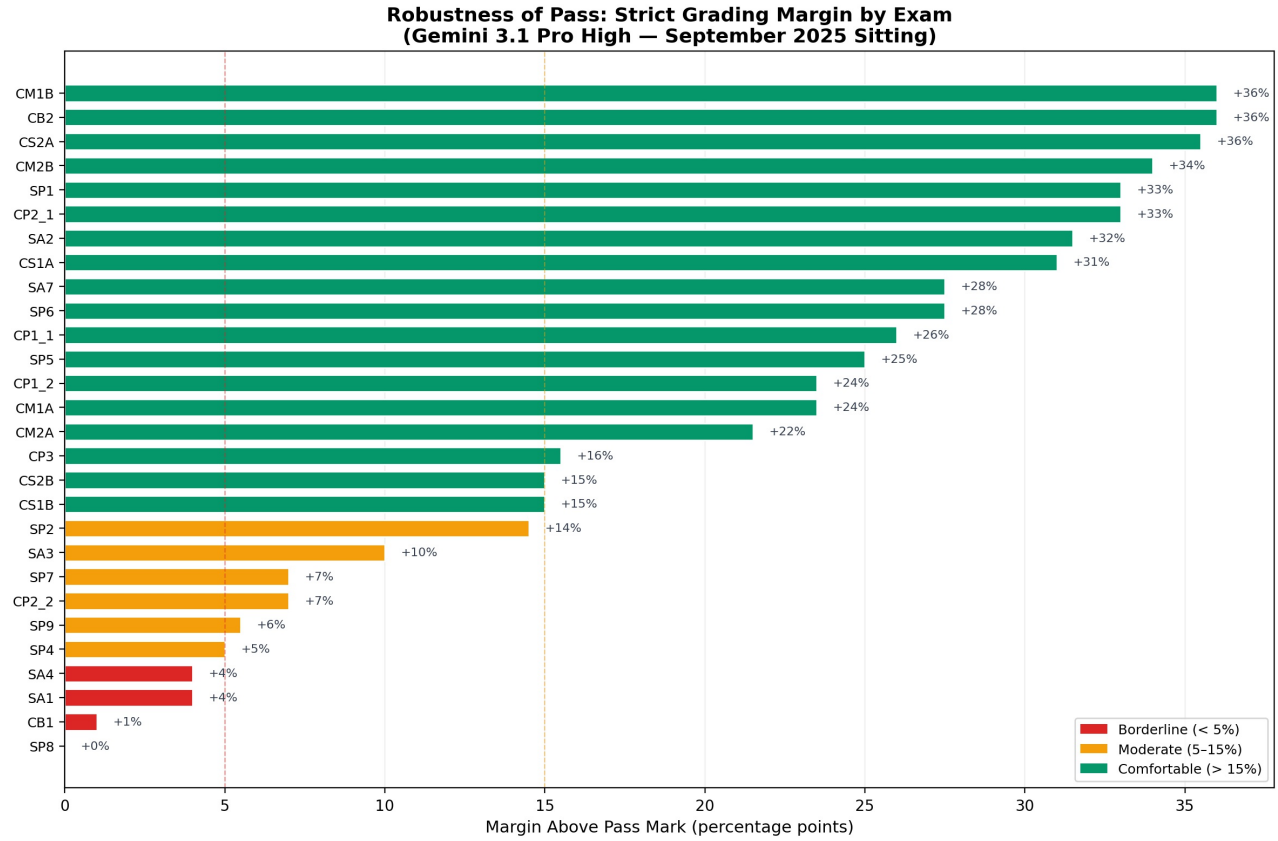
"Strict Margin" = Strict score minus pass mark.

3.3 Robustness Analysis

While the headline 100% pass rate holds across all three grading conditions, the margins under Strict grading reveal varying degrees of confidence:

- **Comfortable passes (Strict margin > 15%):** 16 of 28 exams (57%). These results are robust — the model would pass even with significant downward adjustment.
- **Moderate passes (Strict margin 5–15%):** 7 exams (25%) — CP2_2, CP3, CS1B, CS2B, SA3, SP2, SP7.
- **Borderline passes (Strict margin < 5%):** 5 exams (18%) — **SP8 (+0.0%)**, CB1 (+1.0%), SA1 (+4.0%), SA4 (+4.0%), SP4 (+5.0%).

SP8 is the most precarious result: the Strict score of 58.0% exactly equals the 58% pass mark. A single mark deducted would change this to a fail.



3.4 The Role of the Strict Marker as a Stress Test

Marking 28 exam papers by hand would be impractical at scale, so the grading was performed by the same AI model. To reduce reliance on the model's own judgement about its answers, the Strict persona was designed to be deliberately conservative — a stress test rather than a realistic marking standard. By awarding marks only where answers exactly match the examiner's report in wording and method, the Strict scores provide a lower bound: if the model passes even under this level of scrutiny, we can be confident the material knowledge is there.

Spot-checking against a sample of grade reports confirmed that the Strict persona was indeed overly harsh in practice. The disagreements between Strict and Fair scores fell into four consistent patterns:

1. Alternative valid approaches. In CS1A Question 6(iii), the LLM used the standard Beta distribution mode formula to compute a Bayesian estimator, obtaining the correct numerical answer. The Strict marker awarded 1.5/3 for using a "standardised shortcut" rather than differentiating from first principles. The Fair marker awarded 3/3, recognising that applying a known formula correctly demonstrates competency.

2. Notation and parameterisation differences. In CM2A Question 6(ii), a PDF-to-Markdown extraction artefact caused the LLM to work with $U(w) = w - w^2/2d$ instead of $U(w) = w - dw^2$. The derivation and final answer (66,469.75) were mathematically identical. The Strict marker awarded 0/2 for using "the wrong parameterisation"; the Fair marker awarded 2/2 for correct reasoning.

3. Presentation style. In CM2B Question 2(v), the LLM explained CIR model dynamics in accessible language ("forces the short rate into a narrower corridor"). The Strict marker deducted 2 marks for "slightly too accessible verbiage" instead of formal stochastic definitions. The Fair marker awarded 3/4, noting the answer reflected "standard actuarial conceptual expectations."

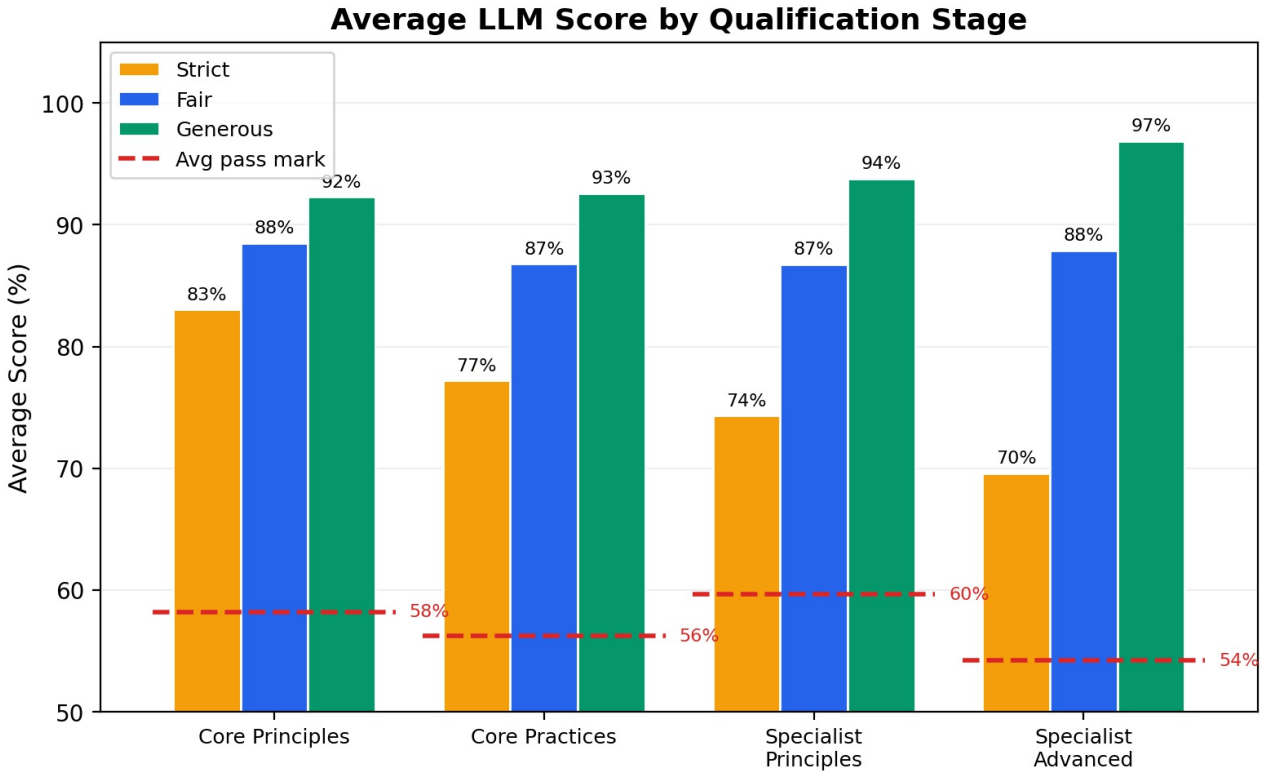
4. Keyword matching. In SA1 Question 2(ii), the LLM gave a correct definition of immediate-need LTC products in slightly different words to the marking scheme. The Strict marker awarded 2/3 despite acknowledging the answer was "very close" to the scheme text. The Fair marker awarded 3/3.

These examples confirm the Strict persona's role as a conservative floor. Where the two scores converge — predominantly on mathematical papers with objectively right-or-wrong answers — the grading is straightforward and both personas agree. Where they diverge, the Strict score reflects literal keyword-matching against the examiner's report rather than any genuine deficiency in the answer. The Fair score, which exercises the kind of professional judgement a human marker might apply, is the primary metric throughout this paper.

3.5 Performance by Subject Area

- **CB (Business):** Scores ranged from 61.0% to 94.0% (Strict). CB2 was a standout (94% Strict), while CB1 was borderline under Strict grading. The model generated valid business arguments but was penalised under Strict marking for not using specific keywords from the mark scheme.
- **CM (Mathematics):** Consistently strong, with Strict scores of 80.5%–95.0%. CM1B achieved a perfect 100% under Fair grading. The model navigated multi-state processes, yield curve derivations, and joint-life assurance calculations with precision.
- **CS (Statistics):** Strong on Paper A (89.0%–91.5% Strict), with lower but passing scores on Paper B (71.0%–73.0% Strict) where data interaction was required.
- **CP (Practice):** Variable performance. CP2 Paper 1 (modelling) scored 90.0% Strict, while CP2 Paper 2 (communication) was the weakest Fair result at 72.5%. The model produced readable summary documents (CP3) and replaced iterative Excel methods with algebraic solutions.
- **SP (Specialist Principles):** Wide range from SP8 (58.0% Strict, borderline) to SP1 (93.0% Strict). Performance correlated with the balance of calculation vs. discursive content.

- **SA (Specialist Advanced):** These long-form, scenario-based papers showed the widest grading variance. The model demonstrated strong commercial reasoning but sometimes favoured complex systemic explanations over the specific points targeted by examiners.



3.6 Performance by Question Type

Question Type	Typical Performance	Notes
Multiple choice	Moderate to High	Strong knowledge base; occasionally over-analyses simple questions
Short calculation	Very High	Precise formula application and accurate results
Long calculation	High	Sound methodology; occasional transcription or magnitude errors
Bookwork / definitions	High	Accurate but sometimes generic rather than exam-specific
Discussion / essay	High (with variance)	Strong commercial insight; can miss specific examiner bullet points
Mathematical proofs	Very High	Rigorous derivations with clear logical steps
R/Excel practical	High	Actual code execution; results are computed, not estimated

4. Analysis and Observations

4.1 Mathematical Reasoning

The model's strongest area was mathematical reasoning. It consistently produced correct derivations for complex actuarial calculations, including multi-state transition models, forward rate structures, joint-life assurance functions, and unit-linked contract valuations. It routinely avoided common errors noted in the examiner reports as pitfalls for human candidates — for example, correctly treating continuous forward rates, handling deferment periods in joint-life calculations, and applying zeroisation procedures.

4.2 Commercial and Discursive Performance

In essay-style and discussion questions, the model produced well-structured responses with relevant commercial perspective. It identified economic dynamics such as cross-subsidies, clientele effects, and regulatory constraints. However, it sometimes provided broader, more comprehensive answers than the marking scheme required, leading to marks being lost under Strict grading where examiners expected specific, concise bullet points rather than expansive analysis. This "textbook penalty" was most pronounced in Specialist and Advanced papers.

4.3 Practical Assessment

The model's ability to execute Python and R code was a significant factor in its performance. Rather than providing theoretical derivations, the model wrote and ran computation scripts against the provided data files, producing exact numerical results. In some cases, it replaced iterative Excel methods (such as Goal Seek) with direct algebraic solutions — a valid approach that typically earned full marks under Fair grading.

The strongest practical paper result was CM1B (95% Strict, 100% Fair), demonstrating that with appropriate tooling, the model can handle complex actuarial spreadsheet calculations end-to-end.

4.4 Grading Calibration

The tri-persona framework serves two purposes: providing a plausible range for the "true" score, and making the grading agent's reasoning auditable. As discussed in Section 3.4, the most informative comparison is between the Strict and Fair personas, since the Generous persona is ceiling-bounded and adds limited signal.

The Strict persona proved useful as a stress test: it identified the five borderline exams (SP8, CB1, SA1, SA4, SP4) that would most benefit from independent human re-grading. However, its tendency to penalise valid alternative approaches, correct-but-differently-worded answers, and standard formula shortcuts means it likely *understates* the model's true performance on discursive papers.

The Fair persona is intended as the primary metric, approximating a human marker exercising professional judgement. While the self-grading concern (Section 5) applies, the concrete examples in Section 3.4 suggest the Fair marker's reasoning is defensible — it awards marks for mathematically equivalent work and correct-but-differently-expressed answers, which is consistent with how IFoA markers are instructed to apply professional judgement.

5. Limitations

This study has several important limitations that should be considered when interpreting the results:

1. **Self-grading bias.** The same model family (Gemini 3.1 Pro) was used for both exam-taking and grading. While the two phases used independent agent sessions with no shared context, LLMs have been shown to exhibit self-preference in evaluation tasks. The tri-persona framework and selective human spot-checking partially mitigate this, but do not eliminate it. Independent human re-grading of the full set would provide a more reliable assessment.

2. **Single model, single sitting, single run.** The experiment evaluated one model on one exam sitting, with each exam attempted once. No claims of statistical significance can be made from n=1. The results may not generalise to other sittings (which test different scenarios), other models, or repeat runs (LLM outputs are stochastic and may vary between attempts).

3. **PDF-to-Markdown conversion quality.** Mathematical notation, complex tables, and formatted formulae can be garbled during PDF extraction. This may have affected the model's interpretation of some questions and the grader's interpretation of some marking scheme points, particularly in papers heavy with mathematical notation.

4. **Variable pass marks.** Pass marks are set by the IFoA per sitting and ranged from 45% (SA2) to 64% (SP5, SP7) in this sitting. A "pass" on SA2 represents a lower absolute bar than a "pass" on SP5. The 100% pass rate should be read in conjunction with the individual margins shown in Section 3.2.

5. **Grading persona calibration.** The Strict persona penalises valid alternative approaches, equivalent notation, and presentation style (see Section 3.4), likely understating the model's true performance. The Generous persona is ceiling-bounded at 100% and adds limited signal. The Fair score, used as the primary metric, is a single point estimate between these bounds. On discursive papers the gap between Strict and Fair can exceed 20 percentage points, reflecting the inherent subjectivity of marking essay-style answers rather than a failing of the model.

6. **Reproducibility.** The experiment was not re-run to assess consistency. Future work should include multiple runs to establish score distributions and confidence intervals.

6. Discussion: Implications for the Profession

The results provide strong evidence that a modern LLM can meet the technical standard of the IFoA examination syllabus, subject to the limitations discussed above. But passing exams and replacing actuaries are very different propositions. Understanding why requires looking at what the profession demands beyond examinations.

6.1 Exams Are One Layer of a Much Deeper System

The IFoA's framework for ensuring quality extends well beyond the examination syllabus. Exams are the most visible layer, but they sit within a system of at least ten interlocking requirements designed to ensure that actuarial work meets the standard the public and regulators expect. An AI that passes all 28 exam papers has demonstrated capability in only one of these layers.

Qualification requires supervised work experience. Before an IFoA student can qualify as an Associate or Fellow, they must complete the Personal and Professional Development (PPD) requirements: a minimum of 24 months of supervised workplace experience (for Associate) with additional time to Fellowship. PPD assesses over 30 competencies across three objectives — effective communication, problem solving and decision making, and professionalism — with mandatory competencies (such as presenting actuarial work to non-technical audiences, and understanding the role of peer review) that must be signed off by a workplace supervisor. The PPD framework was designed for human students and currently has no mechanism for an AI system to participate — but the underlying activities (working on real actuarial problems, receiving feedback, demonstrating competencies) are not inherently beyond what an AI could do.

Practising Certificates gate the most important roles. Certain reserved regulatory roles — Scheme Actuary, With-Profits Actuary, Chief Actuary (Life and Non-Life) — require a Practising Certificate (PC). A PC requires Fellowship plus demonstrated competency against a detailed role-specific framework covering personal skills, technical knowledge, and practical experience. Applicants must describe in their own words how they meet each competency; certificates must be renewed periodically. These roles carry personal regulatory accountability that cannot be delegated to a machine.

The Actuaries' Code imposes ethical obligations. All IFoA members are bound by the Actuaries' Code, which sets six principles: integrity, competence and care, impartiality, compliance, speaking up, and communication. Several of these — particularly speaking up against unethical practices, exercising impartial professional judgement free from conflicts of interest, and acting with integrity — are fundamentally human obligations that require moral agency and personal accountability.

Peer review is mandatory. Under Actuarial Profession Standard X2 (APS X2), IFoA members must consider and apply work review, including independent peer review, to all actuarial work. The level of review required depends on the difficulty of the work, the extent of judgement applied, and the financial significance of the consequences. Independent peer review must be conducted by someone with sufficient experience to have taken responsibility for the work themselves.

Continuing professional development is required indefinitely. Qualified actuaries must complete a minimum of 15 hours of Continuing Professional Development (CPD) per year, including at least 2 hours focused on professional and ethical skills. Members participate in Reflective Practice Discussions and may be called upon by the IFoA to evidence their CPD record. This is a career-long obligation, not a one-off hurdle.

Organisational quality assurance adds a further layer. The IFoA's Quality Assurance Scheme (QAS) accredits organisations — not just individuals — assessing whether their policies and working culture support quality actuarial work, manage conflicts of interest, and foster a speaking-up culture. Accreditation involves independent assessment through qualitative interviews and lasts six years with interim monitoring.

External oversight and regulation sit above the profession itself. The Financial Reporting Council (FRC) provides independent oversight of the IFoA, sets Technical Actuarial Standards (TAS 100, TAS 200, TAS 300), and operates an Actuarial Monitoring Programme that reviews the quality of technical actuarial work in practice. The Prudential Regulation Authority (PRA) and Financial Conduct Authority (FCA) impose additional requirements on actuaries in regulated roles. And the IFoA's own Disciplinary Scheme can investigate misconduct, impose fines, require retraining, or exclude members for up to five years — sanctions that require a person to apply them to.

In short: the profession has never treated passing exams as proof that someone is ready to sign off work alone. The exams are the entry point to a system of supervised experience, ethical obligations, peer review, ongoing development, organisational standards, and external regulation. Every one of these layers exists because exam-level competence is necessary but not sufficient.

6.2 Passing Is Not the Same as Being Right

This point is worth making concrete. Under strict marking, five exams in this experiment passed with a margin of less than 5 percentage points. On the borderline papers, the model was arguably wrong or incomplete on 35–40% of the content. Would a client, regulator, or board want actuarial advice that is 60% right? Almost certainly not.

But then the pass marks for these exams ranged from 45% to 64%. The profession does not consider a newly qualified actuary who has just cleared the pass mark ready to work unsupervised either — that is precisely why the layers described in Section 6.1 exist. The parallel holds: an LLM that passes exams sits in a similar position to a newly qualified actuary. It can produce useful technical work, but it is not trustworthy as the sole author of advice that carries professional responsibility.

6.3 AI as a Team Member, Not a Replacement

The more practical question is not whether AI can replace actuaries, but whether it can contribute as a team member within the existing professional frameworks described above. Two applications follow naturally from the results:

AI-assisted drafting. The model's strong performance on technical calculations (CM1B: 100% Fair) and structured written analysis suggests it could produce initial drafts of actuarial advice, reports, or model documentation — work that a qualified actuary would then review, challenge, and refine. This is analogous to how junior team members currently contribute: they do the groundwork, and experienced professionals ensure the quality. The efficiency gain comes from the speed of the initial draft; the quality assurance comes from the human review layer that APS X2 already mandates.

AI-assisted peer review. Conversely, the model's ability to evaluate work against a marking scheme (as demonstrated by the grading phase of this experiment) suggests a role in quality assurance — an AI peer reviewer that sits alongside human peer review, not replacing it. A model that can identify mathematical errors, check derivations against standard formulae, and flag inconsistencies in assumptions could catch issues that a time-pressed human reviewer might miss. The model's tendency to apply marking criteria literally (as shown by the Strict persona) could be an advantage in this context, where thoroughness matters more than professional judgement.

Model documentation and audit trails. In the CP2 papers, the model built complete Excel models with parameter sheets and automated checks. The same capability could be applied to writing methodology documentation, model specifications, and audit trails — work that is time-consuming but essential, particularly given TAS 100 requirements around documentation of technical actuarial work. An AI-generated first draft, reviewed and refined by the responsible actuary, could significantly reduce the documentation burden.

Assumption-setting research. The model demonstrated strong ability to synthesise information and identify relevant factors across the SA papers. It could compile research to support assumption-setting processes — gathering market data, summarising regulatory guidance, identifying precedents and peer benchmarks — for an actuary to then apply professional judgement to. The research leg-work is often the most time-intensive part of the process.

Code review and validation. The model wrote and executed actuarial calculation scripts across practical exams (CM1B, CM2B, CS1B, CS2B). The same capability could be turned to reviewing existing actuarial code — R, Python, VBA — for mathematical errors, logic issues, or inconsistencies with stated methodology. This is a natural extension of the Strict marking persona's literal checking approach, and code review is an area where thoroughness and pattern-matching are more valuable than professional judgement.

Regulatory horizon scanning. The model's ability to process large volumes of text and extract relevant points (demonstrated across the SP papers) suggests a role in monitoring regulatory changes, summarising consultation papers, and flagging implications for existing work.

Training and exam preparation. The model could generate practice questions, work through solutions with students, and provide personalised feedback — essentially acting as a study partner for actuarial students. This is arguably the most immediately deployable application, since there are no professional accountability concerns: the student is learning, not producing work that others will rely on.

Client communication drafts. CP3 tested the model's ability to produce clear written summaries of technical work for non-technical audiences — a daily task for many actuarial teams. Initial drafts of client reports, board papers, or policyholder communications could save significant time, with the responsible actuary ensuring accuracy, tone, and appropriateness for the audience.

None of these applications replace the qualified human who carries the professional responsibility. All work within the existing framework of review, sign-off, and accountability.

6.4 Could an AI Meet the Broader Professional Requirements?

It is worth considering — even speculatively — whether the non-exam requirements could ever be adapted for AI systems. Could an AI demonstrate the PPD competencies? Many of them — applying technical knowledge to workplace problems, communicating results clearly, solving multi-step analytical problems — look feasible given the capabilities demonstrated in this experiment. Others, such as adapting a presentation based on a client's body language or navigating the politics of a difficult board meeting, remain beyond current AI systems. Could an AI hold a Practising Certificate? The competency frameworks require personal accountability and regulatory responsibility that currently presuppose a legal person — but this is a legal and regulatory question, not a technical one. Could an AI be disciplined? The IFoA's Disciplinary Scheme imposes sanctions — fines, retraining, exclusion — on individuals, which has no direct analogue for a software system, though analogous concepts (revoking access, mandating retraining/fine-tuning, version withdrawal) are not hard to imagine.

The deeper question is one of accountability. The profession's quality framework is built on the principle that actuarial work carries consequences for the public, and that identifiable people must be personally accountable for its quality. Whether and how that framework evolves to accommodate AI tools — and who bears responsibility when an AI team member contributes to a piece of work — is a question for the profession, its regulators, and the public it serves.

6.5 Actuarial Education and Lessons from Other Professions

The actuarial exams are not the first professional examinations to be passed by AI. GPT-4 passed the Uniform Bar Exam at approximately the 90th percentile, the US Medical Licensing Examination (USMLE) Steps 1–3 with accuracy rates of 92–95%, and sections of the CPA exam. The responses from those professions offer useful precedent.

The legal profession has moved quickly. The American Bar Association issued Formal Opinion 512 in July 2024 on lawyers' use of generative AI, and dozens of state bars have since published their own guidance. Pennsylvania now mandates explicit disclosure of AI use in all court submissions. New York requires at least two annual Continuing Legal Education credits in AI competency. Perhaps most significantly, the bar exam itself is being redesigned: by 2028, at least half of US states are expected to adopt the NextGen Bar Exam, which shifts toward skills-based assessment rather than knowledge recall. In medicine, California enacted Assembly Bill 3030 in early 2025, requiring healthcare providers to disclose when generative AI is used in clinical communications.

The common threads across these professions are instructive:

- **Transparency requirements** — mandating disclosure when AI has contributed to professional work.
- **AI competency as a professional obligation** — adding AI literacy to continuing education requirements, on the basis that professionals need to understand the tools they (or their colleagues) are using.
- **Exam redesign toward skills-based assessment** — shifting away from knowledge recall and toward the application of judgement, since recall is precisely what AI does well.
- **Integration rather than prohibition** — no major profession has attempted to ban AI use; the focus has been on governance, disclosure, and quality frameworks.

The IFoA's current exams heavily weight technical derivation and knowledge recall — capabilities where LLMs perform strongly. If an AI can reliably derive a joint-life assurance function or explain the dynamics of a CIR model, the exam's value shifts from testing whether a candidate *can* do these things to testing whether they know *when* to do them, *why*, and what the limitations are. Professional judgement, ethical reasoning, and the critical evaluation of model outputs (including AI-generated outputs) are harder to automate and may deserve greater weight in future syllabi. The legal profession's move toward the NextGen Bar Exam suggests this kind of evolution is both feasible and, increasingly, expected.

6.6 Tooling Integration

The model's strong practical paper performance (enabled by code execution capabilities) demonstrates that the practical barrier is not reasoning ability but tooling access. As LLMs become integrated with enterprise environments — running Python, R, and Excel against live datasets — the distinction between "theoretical" and "practical" AI capability narrows. The model did not merely describe how to build a unit-linked projection; it wrote the code, ran it, and produced the numbers. Actuarial teams already use sophisticated tooling; the question is whether AI becomes part of that toolkit.

6.7 The Profession Is Already Engaging with AI

It is worth noting that the actuarial profession is not waiting passively for AI to arrive. The IFoA's AI Ethics Working Party has published a series of practical articles addressing the challenges head-on, including guidance on managing LLM hallucination ("After the headlines: six practical techniques for managing LLM hallucination", December 2025), an ethical checklist for developing AI models (January 2026), and a framework for understanding and controlling LLM use in practice ("Thinking fast and slow", November 2025). The working party has also made the case that actuaries are uniquely positioned to lead AI governance, given their existing expertise in modelling, uncertainty, and risk.

This body of work is directly relevant to the results presented here. An AI system that passes exams but occasionally hallucinates — producing plausible but incorrect reasoning — is precisely the kind of tool that requires the governance frameworks the working party describes. The ethical checklist for AI model development, for instance, provides a ready-made framework for organisations considering how to deploy LLM-generated actuarial work within their existing quality assurance processes. The profession's proactive engagement with these questions is a strength, and positions it well to integrate AI tools thoughtfully rather than reactively.

7. Conclusion

This experiment provides strong evidence that a modern Large Language Model can pass the IFoA actuarial examinations across all four qualification stages. The 100% pass rate achieved by Gemini 3.1 Pro High across 28 papers, with an average Fair score of 87.6%, demonstrates substantial capability in mathematical reasoning, statistical analysis, and commercial judgement.

However, the results come with material caveats: the use of self-grading by the same model, the absence of full independent human validation, borderline Strict passes on several papers, and the inherent limitations of a single-sitting, single-run experiment. The findings should be treated as indicative rather than definitive.

Passing exams is not the same as being a competent actuary. The IFoA's qualification framework requires supervised work experience, ethical obligations, peer review, ongoing professional development, organisational quality assurance, and external regulatory oversight — none of which an AI system can currently satisfy. These layers exist because the profession has always recognised that exam-level competence is necessary but not sufficient for work that affects the public interest.

The same professional frameworks that protect quality today provide a natural starting point for integrating AI into actuarial teams — not as a replacement for qualified professionals, but as a capable team member whose output is reviewed, challenged, and improved by the people who carry the professional responsibility.

Actuaries are, in many ways, better placed than most professions to lead this transition. The core skills of the profession — quantifying uncertainty, understanding model limitations, communicating risk to non-technical audiences, and building governance frameworks around complex analytical tools — are precisely the skills that the responsible deployment of AI demands. The question is not whether AI will play a role in actuarial work, but whether the profession chooses to lead that integration or respond to it. The depth of the IFoA's existing quality framework, and the proactive work of groups like the AI Ethics Working Party, suggest the foundations for leadership are already in place.

This research forms part of an open-source initiative evaluating AI capabilities in actuarial science. Full methodology, grade reports, and raw data are available in the project repository.

Experiment conducted: February–March 2026.